(A.) To prove: assigning each point to the nearest centroid minimizes the k-means objective given fixed centroids.

Objective Function :-

$$J = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - u_k\|^2$$

x_i : data point
u_k : Centroid for cluster C_k.

→ For fixed centroids u_k, each data point x_i is clustered with the nearest centroid.

→ Focusing on a single data point x_i, we minimize,

$$\|x_i - u_k\|^2 \; ; \quad k \in \{1, \ldots, K\}.$$

→ Hence, the optimal assignment of x_i is to the k using the "minimum squared Euclidean distance".

→ Assigning each point to the nearest centroid minimizes the contribution of that point to the total objective.

(B) To Prove: Given fixed memberships, the mean of the points in a cluster minimizes the objective function.

→ let's fix cluster C_k (with assigned members) and find the optimal centroid u_k that minimizes:

$$J_k = \sum_{x_k \in C_k} \| x_i - u_k \|^2$$

→ Minimizing over u_k, derivating it w.r.t. u_k.

$$\frac{dJ_k}{du_k} = \frac{d}{du_k} \sum_{x_k \in C_k} \| x_i - u_k \|^2$$

$$\Rightarrow \sum_{x_k \in C_k} -2(x_i - u_k) = 0$$

$$\Rightarrow \sum_{x_k \in C_k} (x_i - u_k) = 0$$

$$\Rightarrow u_k = \frac{1}{|C_k|} \sum_{x_k \in C_k} x_i$$

→ The mean of cluster members minimizes the sum of squared ~~errors~~ distances to the centroid.

(c) The convergence of k-means is guaranteed due to a combination of monotonicity, finite configuration space, and deterministic updates even though it may settle at a local minimum.

- For n data points and k clusters, there are only a ~~distinct~~ finite number of distinct partionings. This prevents infinite oscillation between configurations.

- Since each iteration produces either a lower or equal objective value and no two different clustering can yield the exact same objective unless they're identical, the algorithm cannot cycle indefinitely.

- The objective function value decreases for any change in assignment or centroid. Due to this, the algorithm can't bounce around between different suboptimal points.

- Given fixed initial centroids, the membership and centroid updates follow a deterministic path, ensuring that the process doesn't randomly explore new configurations. It's always moving in a directed path towards a stationary point.

- Why it might not reach Global minimum:
  - Initial centroid choice may trap the algorithm.
  - Hence, random restarts or smarter initialization like k-Means++ are often used to find better solutions.